Milestone - 4 Report

Introduction:
This milestone includes using larger models like DaBERTa fine-tuning. This was done in multiple approaches, varying the random parameters and splitting strategies. The experiments were as follows:

- 3 epochs (DaBERTa + RoBERTa)
- 8 epochs fine tuning
- 16 epochs fine tuning
- 2 stage tuning: 8 epochs (2e-5)+ (5-e6)
- 2 stage: 8 epochs (2e-5) + 5 epochs (5-e6)

Performance:

Model	Validation Loss	Prediction performance (MAP@3)
Baseline (sample_submission.csv)	-	0.3525
TF-IDF	0.29617	0.32294
Word2vec	0.34792	0.30423
sBERT	0.42308	0.38653
BART-MNLI	0.56583	0.49985
DaBERTa (3 epochs)	-	0.62177
RoBERTa (3 epochs)	-	0.49958
DaBERTa (8 epochs at 2e-5)	-	0.74064
DaBERTa (16 epochs at 2e-5)	-	0.71945
DaBERTa (8(2e-5) + 8(5e-6))	-	0.73316
DaBERTa (8(2e-5) + 5(5e-6))	-	0.74397

Conclusion:
The experiments began with simple 3 epoch training runs, between DaBERTa and RoBERTa. Since DaBERTa seemed like a natural better fit, it was chosen for further training runs. It was performed using 8 epochs, which produced a very desirable result of 0.74, it was taken further to 16 runs which largely overfit on the data and produced a lower score than the 8 epochs. Then the next experiment was run using 8 normal + 8 additional epochs at a lower learning rate run, which produced a better result than overfitting 16 epochs but worse than 8 epochs initially, But

the finetuning approach seemed to have produced better loss curves and accuracies even in the split validation data, so an additional experiment with 8 normal epochs and 5 additional epochs which improved on the score from 8 epochs, this was the point where the model was not overfitting anymore.